

Phase 2 report – Linear-response $H(\omega)$ (H2 part A)

Verdict: **PASS**

Hypothesis (H2 part A)

For the negative-loop (E-I) archetype with default FCS weights ($w_{XA} = 11$, $w_{AI} = 11$, $w_{IA} = -11$, $p_{\text{thin}} = 0.7$), the single-pole low-pass approximation of the linear-response transfer function

$$H_{i(\omega)} = \frac{\partial \frac{\Phi_i}{\partial} \mu_i}{1 + i\omega\tau_m}$$

at the Siegert fixed point, plugged into the closed-loop matrix $M(\omega) = I - H(\omega)J$, satisfies the $\omega = 0$ self-consistency: the time-domain Jacobian $A = (1/\tau_m)(-I + \text{diag}(g)J)$ has the same spectrum as $(\text{diag}(g)J - I)/\tau_m$ obtained from $\det M(0) = 0$.

Operating point

Calibration locked in from Phase 1:

- $\alpha = 0.2500$, $\beta = 0.004292$, $\tau_m = 2.3504$, $\tau_{\text{ref}} = 0.3611$.

Topology and effective Jacobian:

$$J_{\text{FCS}} = \begin{pmatrix} 0 & -11 \\ 11 & 0 \end{pmatrix}, \quad J_{\text{eff}} = \alpha J_{\text{FCS}} = \begin{pmatrix} 0.0000 & -2.7500 \\ 2.7500 & 0.0000 \end{pmatrix}$$

Self-consistent Siegert fixed point:

- $\nu^* = (0.3520, 0.1793)$
- $\mu^* = (1.4319, 0.9680)$
- $\sigma^* = (0.4307, 0.3442)$

DC gains:

- $\partial\Phi_A/\partial\mu_A = 0.3163$
- $\partial\Phi_I/\partial\mu_I = 0.3598$

Time-domain Jacobian eigenvalues

Spectrum of $A = (1/\tau_m)(-I + \text{diag}(g)J_{\text{eff}})$:

$-0.4255 + 0.3947i$, $-0.4255 - 0.3947i$

Maximum real part: -0.4255 (negative => stable focus or node; positive => unstable). Maximum |imaginary| part: 0.3947 (non-zero => oscillatory dynamics).

Self-consistency

Closed-loop characteristic polynomial $\det M(\omega) = \det(I - H(\omega)J)$ at $\omega = 0$ has roots $1 - \tau_m\lambda$ in correspondence with the Jacobian spectrum. Numerical residual (max difference between sorted spectra):

$$|\text{residual}| = 0.00e + 00$$

Gate $\leq 10^{-3}$: **PASS**.

Imaginary-axis crossings of $\det M(\omega)$ in $[0.01, 30]$ rad/time-unit: none in tested range.

Bode plot of closed-loop response

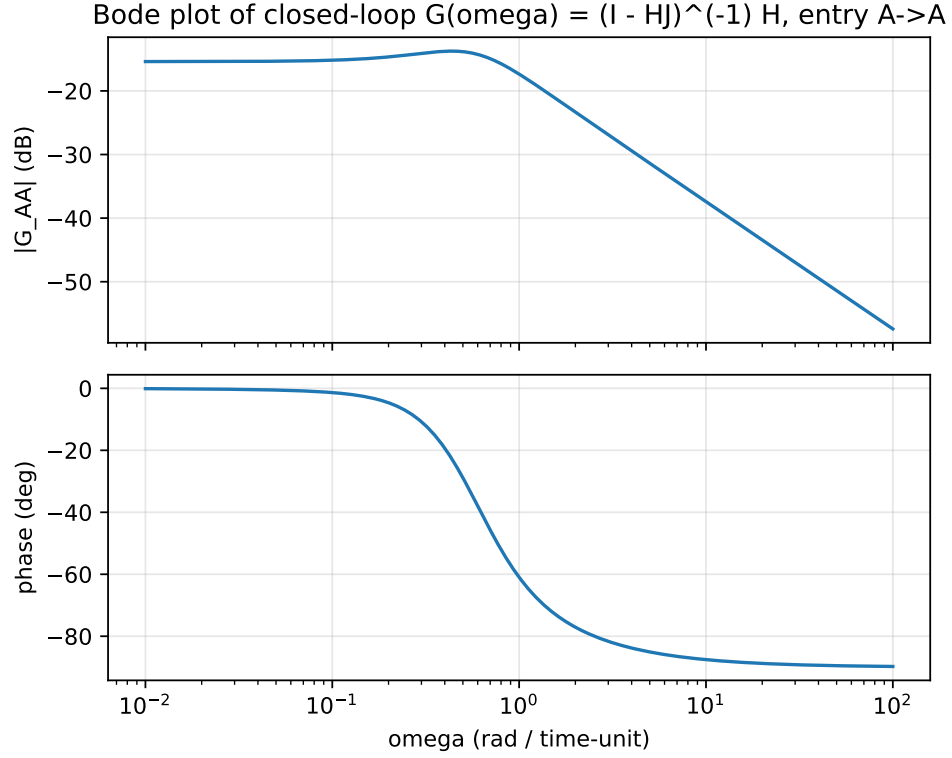


Figure 1: Bode magnitude (top) and phase (bottom) of the closed-loop transfer $G(\omega) = (I - H(\omega)J)^{-1}H(\omega)$, entry A $\rightarrow$ A. The single-pole low-pass shape is consistent with a stable configuration where the negative loop adds bandwidth-limited damping.

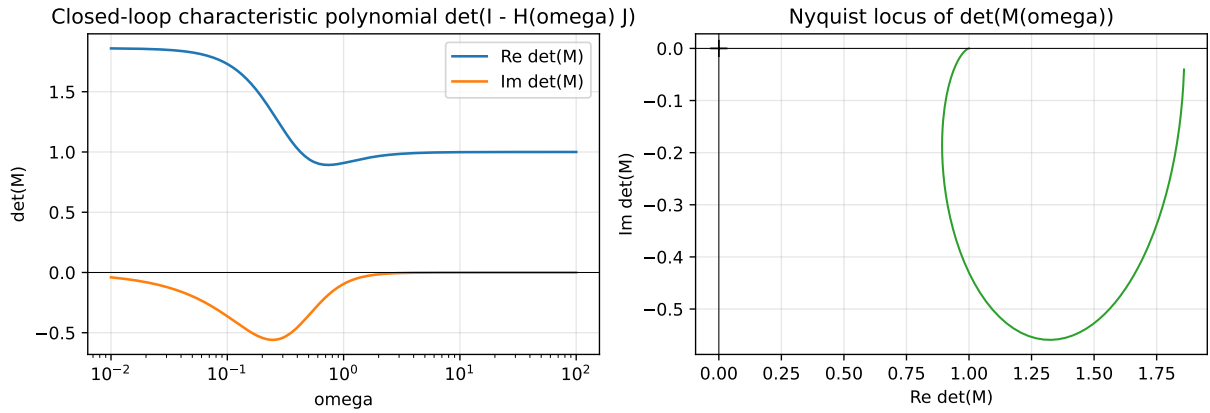


Figure 2: Left: real and imaginary parts of $\det M(\omega)$. Right: Nyquist locus of $\det M(\omega)$. The locus does not encircle the origin, consistent with the Jacobian eigenvalues having negative real parts.

Overall verdict

PASS.

The Phase 2 self-consistency check confirms the closed-loop machinery correctly reduces to the time-domain Jacobian at $\omega = 0$. Phase 3 (H2 part B) extends to non-zero ω : comparing predicted spectral peaks against LI&F-population FFT measurements.